

SDG Goal 8 Decent work and economic growth

SDG Target 8.4 Improve progressively, through 2030, global resource efficiency in consumption and production and endeavour to decouple economic growth from environmental degradation, in accordance with the 10-Year Framework of Programmes on Sustainable Consumption and Production, with developed countries taking the lead

SDG Indicator 8.4.2 Domestic material consumption, domestic material consumption per capita, and domestic material consumption per GDP

Time series Domestic material consumption (DMC)

1. General information on the time series

- Date of national metadata: June 9, 2022
- National data: <https://sdg-indicators.de/8-4-2>
- Definition: The time series measures the amount of materials used in a national economy in metric tons. It is calculated as the domestic extraction of raw materials from the environment plus imports of goods (raw materials, semi-finished and finished products) minus exports of goods. Imports and Exports of goods enter the time series with their actual weight.
- Disaggregation: Not available.

2. Comparability with the UN metadata

- Date of UN metadata: April 2025
- UN metadata: <https://unstats.un.org/sdgs/metadata/files/Metadata-08-04-02.pdf>
- The time series is compliant with the UN metadata.

3. Data description

- The data on domestic material consumption is calculated within the material flow accounting system run by the Environmental-Economic Accounts unit of the Federal Statistical Office. Data on domestic extraction of raw materials stems from various, mostly official, statistics and data sources. For imports and exports, data from foreign trade statistics is used.

For the calculation of population data, the retroactively calculated average population provided by the Federal Statistical Office is used. For 2010, this is based on the 2011 census as well as migration, birth, and death statistics; for the years 2012 to 2021, the calculation is based on the 2022 census.

The data on GDP is calculated by the Federal Statistical Office's National Accounts as a secondary statistic. GDP is adjusted based on a price base changing every year (previous year's price base). After several revisions due to new data input, final results are available four years after the first preliminary release.

4. Access to data source

- Economy-wide material account: GENESIS online 85131-0001:
<https://www-genesis.destatis.de/genesis//online/data?operation=table&code=85131-0001&bypass=true&language=en>
- National accounts – Gross value added, gross domestic product (nominal/price-adjusted): GENESIS online 81000-0001:
<https://www-genesis.destatis.de/genesis//online?operation=table&code=81000-0001&bypass=true&language=en>
- Average population – GENESIS online 12411-0041:
<https://www-genesis.destatis.de/genesis//online?operation=table&code=12411-0041&bypass=true&levelindex=1&levelid=1639396599054#abreadcrumb>
- Population data based on Census 2022 (only available in German):
https://www.destatis.de/DE/Themen/Gesellschaft-Umwelt/Bevoelkerung/Bevoelkerungsstand/_inhalt.html#sprg233540

5. Metadata on source data

- Eurostat metadata on material flow accounts:
https://ec.europa.eu/eurostat/cache/metadata/EN/env_ac_mfa_simsf_de.htm
- Methodology of economy-wide material flow accounting (only available in German):
https://www.destatis.de/DE/Themen/Gesellschaft-Umwelt/Umwelt/UGR/rohstoffe-materialfluesse-wasser/Publikationen/Downloads/gesamtwirtschaftliches-material-konto-fb_5851314189004.pdf
- Quality report: Economic Accounts (only available in German):
<https://www.destatis.de/DE/Methoden/Qualitaet/Qualitaetsberichte/Umwelt/ugr.pdf?>
- Quality Report – Microcensus (only available in German):
<https://www.destatis.de/DE/Methoden/Qualitaet/Qualitaetsberichte/Bevoelkerung/einfuehrung.html>

6. Timeliness and frequency

- Timeliness: t + 24 months
- Frequency: Annual

7. Calculation method

- Unit of measurement: Million tonnes, Tonnes per capita
- Calculation: Not applicable.